Problem 13.6.1

Show that

a) $\sinh z = \sinh x \cos y + i \cosh x \sin y$

b) $\cosh z = \cosh x \cos y + i \sinh x \sin y$

Solution

a)

$$\begin{aligned}
\sinh z &= \frac{1}{2}(e^z - e^{-z}) \\
&= \frac{1}{2}(e^{x+iy} - e^{-x-iy}) \\
&= \frac{1}{2}(e^x e^{iy} - e^{-x} e^{-iy}) \\
&= \frac{1}{2}(e^x(\cos y + i \sin y) - e^{-x}(\cos y - i \sin y)) \\
&= \frac{1}{2}(e^x \cos y + ie^x \sin y - e^{-x} \cos y + ie^{-x} \sin y) \\
&= \frac{1}{2}((e^x - e^{-x}) \cos y + i(e^x + e^{-x}) \sin y) \\
&= \boxed{\sinh x \cos y + i \cosh x \sin y}
\end{aligned}$$

b)

$$\begin{aligned}
\cosh z &= \frac{1}{2}(e^z + e^{-z}) \\
&= \frac{1}{2}(e^{x+iy} + e^{-x-iy}) \\
&= \frac{1}{2}(e^x e^{iy} + e^{-x} e^{-iy}) \\
&= \frac{1}{2}(e^x(\cos y + i \sin y) + e^{-x}(\cos y - i \sin y)) \\
&= \frac{1}{2}(e^x \cos y + ie^x \sin y + e^{-x} \cos y - ie^{-x} \sin y) \\
&= \frac{1}{2}((e^x + e^{-x}) \cos y + i(e^x - e^{-x}) \sin y) \\
&= \boxed{\cosh x \cos y + i \sinh x \sin y}
\end{aligned}$$

Problem 13.6.3

Show that

a) $\cosh^2 z - \sinh^2 z = 1$

b) $\cosh^2 z + \sinh^2 z = \cosh 2z$

Solution

a)

$$\begin{aligned}\cosh^2 z - \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 - \frac{1}{4}(e^z - e^{-z})^2 \\ &= \frac{1}{4}\left[\left(e^{2z} + 2 + e^{-2z}\right) - \left(e^{2z} - 2 + e^{-2z}\right)\right] \\ &= \boxed{1}\end{aligned}$$

b)

$$\begin{aligned}\cosh^2 z + \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 + \frac{1}{4}(e^z - e^{-z})^2 \\ &= \frac{1}{4}\left[\left(e^{2z} + 2 + e^{-2z}\right) + \left(e^{2z} - 2 + e^{-2z}\right)\right] \\ &= \frac{1}{4}(2e^{2z} + 2e^{-2z}) \\ &= \frac{1}{2}(e^{2z} + e^{-2z}) \\ &= \boxed{\cosh 2z}\end{aligned}$$

Problem 13.6.7

Find, in the form $u + iv$

a) $\cos i$

b) $\sin i$

Solution

a)

$$\cos i = \boxed{\cosh 1}$$

b)

$$\sin i = \boxed{i \sinh 1}$$

Problem 13.6.13

Using the definitions, prove

a) $\cos z$ is even, $\cos(-z) = \cos z$

b) $\sin z$ is odd, $\sin(-z) = -\sin z$

Solution

a)

$$\begin{aligned}\cos(-z) &= \frac{1}{2}\left(e^{i\cdot(-z)} + e^{-i\cdot(-z)}\right) \\ &= \frac{1}{2}\left(e^{-iz} + e^{iz}\right) \\ &= \boxed{\cos z}\end{aligned}$$

b)

$$\begin{aligned}\sin(-z) &= \frac{1}{2i}\left(e^{i\cdot(-z)} - e^{-i\cdot(-z)}\right)\\ &= \frac{1}{2i}\left(e^{-iz} - e^{iz}\right)\\ &= \boxed{-\sin z}\end{aligned}$$

Problem 13.6.16

Find all solutions

$$\sin z = 100$$

Solution

$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right) = 100$$

Let $w = e^{iz}$. Then $w - w^{-1} = 100i$

$$w^2 - 200iw - 1 = 0 \quad\implies\quad w = i(100 \pm \sqrt{9999})$$

Thus $e^{iz} = i(100 \pm \sqrt{9999})$, so

$$\begin{aligned}iz &= \ln(i(100 \pm \sqrt{9999}))\\ &= \ln|i(100 \pm \sqrt{9999})| + i\,\mathrm{Arg}(i(100 \pm \sqrt{9999})) + i2\pi n\\ &= \ln(100 \pm \sqrt{9999}) + i\frac{\pi}{2} + i2\pi n\\ &= \pm\ln(100 + \sqrt{9999}) + i\pi\left(2n + \frac{1}{2}\right)\end{aligned}$$

Dividing by i,

$$\boxed{z = \pi\left(2n + \frac{1}{2}\right) \pm \ln(100 + \sqrt{9999}) \quad n \in \mathbb{Z}}$$

Problem 13.7.5

Find

$$\mathrm{Ln}(-11)$$

Solution

$$\begin{aligned}\mathrm{Ln}(-11) &= \ln|-11| + i\,\mathrm{Arg}(-11)\\ &= \boxed{\ln 11 + i\pi}\end{aligned}$$

Problem 13.7.7

Find

$$\mathrm{Ln}(4-4i)$$

Solution

$$\begin{aligned}\mathrm{Ln}(4-4i) &= \ln|4-4i| + i\,\mathrm{Arg}(4-4i)\\ &= \ln\sqrt{4^2+4^2} + i\tan^{-1}\left(\frac{-4}{4}\right)\\ &= \ln\sqrt{32} + i\tan^{-1}(-1)\\ &= \boxed{\frac{1}{2}\ln 32 - i\frac{\pi}{4}}\end{aligned}$$

Problem 13.7.8

Find

$$\mathrm{Ln}(1\pm i)$$

Solution

$$\begin{aligned}\mathrm{Ln}(1\pm i) &= \ln|1\pm i| + i\,\mathrm{Arg}(1\pm i)\\ &= \ln\sqrt{2} + i\tan^{-1}\left(\frac{\pm 1}{1}\right)\\ &= \ln\sqrt{2} + i\tan^{-1}(\pm 1)\\ &= \boxed{\frac{1}{2}\ln 2 \pm i\frac{\pi}{4}}\end{aligned}$$

Problem 13.7.15

Find all values and graph some in the complex plane

$$\ln(e^i)$$

Solution

$$\begin{aligned}\ln(e^i) &= \ln|e^i| + i\,\mathrm{Arg}(e^i) + i2\pi n\\ &= \ln 1 + i + i2\pi n\\ &= \boxed{i(1+2\pi n) \quad n\in\mathbb{Z}}\end{aligned}$$

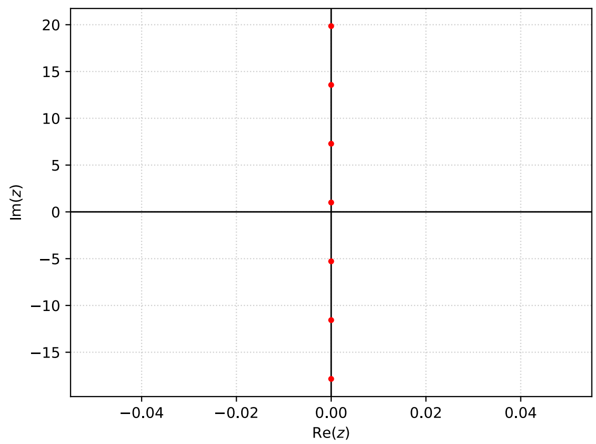

Figure 1: Solutions of $\ln(e^i)$

Problem 13.7.19

Solve for z

$$\ln z = 4 - 3i$$

Solution

Problem 13.7.23

Find the principal value.

$$(1+i)^{1-i}$$

Solution

Problem 14.1.1

Find and sketch the path

$$z(t) = (1 + \frac{1}{2}i)t \quad (2 \leqq t \leqq 5)$$

Solution

Problem 14.1.11

Find a parametric representation and sketch the path

$$\text{Segment from } (-1, 1) \text{ to } (1, 3)$$

Solution

Problem 14.1.19

Find a parametric representation and sketch the path

$$\text{Parabola } y = 1 - \frac{1}{4}x^2 \quad (-2 \leq x \leq 2)$$

Solution

Problem 14.1.21

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Re} z \; dz, \quad \text{C the shortest path from } 1+i \text{ to } 3+3i$$

Solution

Problem 14.1.25

$$\int_C z \exp(z^2) \; dz, \quad \text{C from 1 along the axes to } i$$

Solution

Problem 14.1.29

$$\int_C \operatorname{Im} z^2 \; dz \quad \text{counterclockwise around the triangle with vertices } 0,\ 1,\ i$$

Solution